

Sprint 5 – Building Business Logic with Apex

Placement Management System Submission Implemented Features

- Retrieved Student and Job records using SOQL.
- Prevented duplicate applications.
- Validated eligibility criteria.
- Created and updated Application records.
- Achieved 94% Apex test coverage.

Attached Screenshots :

This report includes screenshots of the **ApplicationService**, **ApplicationServiceTest**, **Apex Test Execution**, and **Code Coverage**. Additional screenshots can be added before final submission.

```
1 public with sharing class ApplicationService {
2
3     public static String submitApplication(Id studentId, Id jobId) {
4         Student__c student = [
5             SELECT Id, Name, CGPA__c, Branch__c, Active_Backlogs__c, Graduation_Year__c
6             FROM Student__c
7             WHERE Id = :studentId
8             LIMIT 1
9         ];
10        Job__c job = [
11            SELECT Id, Name, Minimum_CGPA__c, Eligible_Branch__c,
12                Backlog_Limit__c, Closing_Date__c
13            FROM Job__c
14            WHERE Id = :jobId
15            LIMIT 1
16        ];
17        List<Application__c> existingApplications = [
18            SELECT Id
19            FROM Application__c
20            WHERE Student__c = :studentId
21            AND Job__c = :jobId
22        ];
23        if (existingApplications.isEmpty()) {
24            return 'Duplicate Application! Student already applied.';
25        }
26        if (student.CGPA__c < job.Minimum_CGPA__c) {
27            return 'Application Rejected: Minimum CGPA not satisfied.';
28        }
29        if (student.Branch__c != job.Eligible_Branch__c) {
30            return 'Application Rejected: Branch not eligible.';
31        }
32        if (student.Active_Backlogs__c > job.Backlog_Limit__c) {
33            return 'Application Rejected: Too many active backlogs.';
34        }
35        if (Date.today() > job.Closing_Date__c) {
36            return 'Application Rejected: Application deadline is over.';
37        }
38        Application__c application = new Application__c();
39        application.Student__c = studentId;
40        application.Job__c = jobId;
41        application.Application_Date__c = Date.today();
42    }
43 }
```

```

33 if (student.Branch__c != job.Eligible_Branch__c) {
34     return 'Application Rejected: Branch not eligible.';
35 }
36
37 if (student.Active_Backlogs__c > job.Backlog_Limit__c) {
38     return 'Application Rejected: Too many active backlogs.';
39 }
40
41 if (Date.today() > job.Closing_Date__c) {
42     return 'Application Rejected: Application deadline is over.';
43 }
44
45 Application__c application = new Application__c();
46
47 application.Student__c = studentId;
48 application.Job__c = jobId;
49 application.Application_Date__c = Date.today();
50 application.Status__c = 'Applied';
51
52 try {
53     insert application;
54
55     return 'Application Submitted Successfully';
56 } catch (Exception e) {
57     return e.getMessage();
58 }
59
60 }
61
62 }
63
64
65 public static String updateApplicationStatus(Id applicationId, String status) {
66
67     Application__c app = [
68         SELECT W.Status__c
69         FROM Application__c
70         WHERE Id = :applicationId
71         LIMIT 1
72     ];
73
74     app.Status__c = status;
75
76     update app;
77
78     return 'Application Status Updated Successfully';
79 }
80
81 }

```



SETUP

Apex Classes

```

1  @isTest
2  public class ApplicationServiceTest {
3
4      @isTest
5      static void testSubmitApplication() {
6
7          // Create Student
8          Student__c student = new Student__c(
9              Name = 'Rahul',
10             Email__c = 'rahul@gmail.com',
11             CGPA__c = 8.5,
12             Branch__c = 'CSE',
13             Active_Backlogs__c = 0,
14             Graduation_Year__c = 2027
15         );
16         insert student;
17
18         // Create Job
19         Job__c job = new Job__c(
20             Name = 'Software Engineer',
21             Company__c = 'TCS',
22             Minimum_CGPA__c = 7.5,
23             Eligible_Branch__c = 'CSE',
24             Backlog_Limit__c = 0,
25             Closing_Date__c = Date.today().addDays(10)
26         );
27         insert job;
28
29         Test.startTest();
30
31         String result =
32             ApplicationService.submitApplication(student.Id, job.Id);
33
34         Test.stopTest();
35
36         System.assertEquals(
37             'Application Submitted Successfully',
38             result
39         );
40     }
41
42     @isTest
43     static void testDuplicateApplication() {
44
45         Student__c student = new Student__c(
46             Name = 'Student1',
47             Email__c = 'student1@gmail.com',
48             CGPA__c = 9,
49             Branch__c = 'CSE',
50             Active_Backlogs__c = 0,
51             Graduation_Year__c = 2027
52         );
53         insert student;
54
55         Job__c job = new Job__c(
56             Name = 'Developer',
57             Company__c = 'Infosys',
58             Minimum_CGPA__c = 7,
59             Eligible_Branch__c = 'CSE',
60             Backlog_Limit__c = 0,
61             Closing_Date__c = Date.today().addDays(5)
62         );
63         insert job;
64
65         ApplicationService.submitApplication(student.Id, job.Id);
66
67         String result =

```


Application Test Execution

To run one or more Apex or flow unit tests, click [Select Tests](#). To view the current Apex code coverage for an individual class or your organization, go to the [Apex Classes](#) page.



In Winter '26 and later, this page shows test details for multiple categories of tests, including Apex and flow tests. For more info about how to [Test Your Changes](#), go to [Salesforce Help](#).

[Select Tests...](#)[Developer Console](#)[Options...](#)[View Test History](#)

Abort			
☐	Status	Class	Result
Test Run: 2026-08-03 11:59:46, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☐		[View] ApplicationServiceTest	
Test Run: 2026-08-03 11:34:33, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
		[View] ApplicationServiceTest (7/7) Test Methods Passed	
Test Run: 2026-08-03 11:34:26, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
		[View] ApplicationServiceTest (7/7) Test Methods Passed	
Test Run: 2026-08-03 11:30:10, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
		[View] ApplicationServiceTest (2/2) Test Methods Passed	
Test Run: 2026-08-03 11:29:53, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
		[View] ApplicationServiceTest (2/2) Test Methods Passed	

Code coverage :



Apex Class

ApplicationService

[« Back to List](#)

Apex Class Detail

[Edit](#)[Delete](#)[Download](#)[Security](#)[Show Dependencies](#)

Name ApplicationService

Status Active

Namespace Prefix

Code Coverage 94% (32/34)

Created By [Jaya Sri Padilam](#), 03/08/2026, 10:33 am

Last Modified By [Jaya Sri Padilam](#), 03/08/2026, 11:54 am

[Class Body](#)[Class Summary](#)[Version Settings](#)[Trace Flags](#)

Application Record :

Student

Swathi

New ContactEditNew Opportunity

RelatedDetails

Student Name

Swathi

Owner

Jaya Sri Padilam

Email

swathi@gmail.com

CGPA

8.50

Branch

CSE

Active Backlogs

0

Graduation Year

2027

Created By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Last Modified By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Job record :

Job

Software Engineer

New ContactEditNew Opportunity

RelatedDetails

Job Title

Software Engineer

Owner

Jaya Sri Padilam

Company

TCS

Minimum CGPA

7.50

Closing Date

31/08/2026

Eligible Branch

CSE

Backlog Limit

0

Created By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Last Modified By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Execute Anonymous output :

The screenshot shows the Salesforce Developer Console interface in Google Chrome. The browser address bar displays the URL: `resourceful-narwhal-rffxm7-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage`. The console has several tabs open: `9/07/2026, 10:25:00`, `AppointmentFuture.apxc`, `Log executeAnonymous @8/3/2026, 11:40:21 AM`, and `Log executeAnonymous @8/3/2026, 11:45:05 AM`. The **Execution Log** section is active, showing a table with the following data:

Timestamp	Event	Details
11:45:05:059	USER_DEBUG	[4] DEBUG Duplicate Application! Student already applied.

Below the Execution Log, there are filter options: ☐ This Frame, ☐ Executable, ☒ Debug Only, and ☐ Filter. A link [Click here to filter the log](#) is also present. The **Logs** tab is selected, displaying a table with the following data:

User	Application	Operation	Time	Status	Read	Size
Jaya Sri Padilam	Unknown	/services/data/v67.0/t...	8/3/2026, 11:45:05 AM	Success		6.87 KB